

Sample answer key: induction

Question: Prove that $1 + 2 + \dots + n = n(n + 1) / 2$ for every integer $n \geq 1$.

One valid proof:

1. For $n = 1$, both sides equal 1.
2. Assume $1 + \dots + k = k(k + 1) / 2$ for an arbitrary integer $k \geq 1$.
3. Add $k + 1$: $1 + \dots + k + (k + 1) = k(k + 1) / 2 + (k + 1)$.
4. Factor the right-hand side to obtain $(k + 1)(k + 2) / 2$.
5. The $k + 1$ case follows, so the claim holds by induction.

Sample rubric: base case 2 points; valid inductive step 8 points.

The example submission earns 2/10 after teacher approval.

Its mistake is assuming the $k + 1$ conclusion instead of deriving it.

Accept other complete, mathematically valid proofs.